Canonical Research Designs I: Difference-in-Differences: Extensions and Alternatives

Paul Goldsmith-Pinkham

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Generalizing the Dind approach

- Pivoting slightly: instead of imposing the parallel trends assumption directly through the linear model, we could construct a combination of units to approximate $Y_{it}(0)$
  - This is what one does in the cross-sectional setting with a pscore method! E.g. consider the ATT:

$$\tau_{ATT} = \underbrace{Y(1)}_{\text{Fully observed}} - \underbrace{\hat{Y}(0)}_{\text{Constructed}}$$

- How would one pick? Recall that with p-score methods or regression, weights effectively reweight based on comparability to treated group
  - With panel data, can use pre-treatment data to construct these weights
  - This method is known as synthetic control (and its various descendents)

Synthetic Control example - (Abadie et al. (2010))

- Consider following problem: California bans smoking in 1989. What does that do to smoking?
  - Define estimand: $\tau_{ban,CA} = Y_{california,post}(1) - Y_{california,post}(0)$
  - This is the effect of the *California* smoking ban
  - How can we get at it?

- We need a "synthetic California" as our control

- In an ideal world, the average of the other states would work – however, not clear empirically that they are a good counterfactual

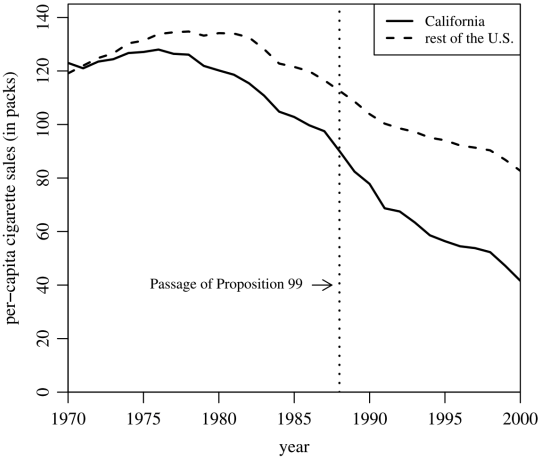

Figure 1. Trends in per-capita cigarette sales: California vs. the rest of the United States.

Generalized setup (Doudchenko and Imbens (2018))

- Consider the following general problem
- We have a panel with T time periods and $N+1$ units. Intervention D_{it} at time T_0 for one unit (unit $i=0$)
- Potential outcomes $Y_{it}(D_{it})$, and we only observe one of the potential outcomes (as per usual)
  - Fundamental problem of causal inference
  - We can also have fixed characteristics X_{it}
- Let $\mathbf{Y}_{a,b}$ denote the vector (or matrix in control case) for $a \in \{\text{treatment}, \text{control}\}$ and $b \in \{\text{pre}, \text{post}\}$ for the treated and control groups in the pre or post period.
- Then, we have observations (analogous setup for the covariates):

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_{t,post} & \mathbf{Y}_{c,post} \\ \mathbf{Y}_{t,pre} & \mathbf{Y}_{c,pre} \end{pmatrix} = \begin{pmatrix} \mathbf{Y}_{t,post}(1) & \mathbf{Y}_{c,post}(0) \\ \mathbf{Y}_{t,pre}(0) & \mathbf{Y}_{c,pre}(0) \end{pmatrix}$$

Generalized panel setup

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_{t,post} & \mathbf{Y}_{c,post} \\ \mathbf{Y}_{t,pre} & \mathbf{Y}_{c,pre} \end{pmatrix} = \begin{pmatrix} \mathbf{Y}_{t,post}(1) & \mathbf{Y}_{c,post}(0) \\ \mathbf{Y}_{t,pre}(0) & \mathbf{Y}_{c,pre}(0) \end{pmatrix}$$

- To estimate $\tau_i = Y_{t,post}(1) - Y_{t,post}(0)$, we need an estimate for $Y_{t,post}(0)$
- What if we just had the cross-section?
  - Note that if D_{it} were randomly assigned, we can derive an estimate using our p-score or regression methods
  - Even without random assignment, one could use covariates to match
  - Our main concern with p-score matching is bias
- Diff-in-diff exploited the panel structure by asserting a particular functional form

$$Y_{it} = \alpha_i + \gamma_t + D_{it}\tau + \epsilon_{it}$$

- Is there something particularly special about this linear additive factor structure?

Generalized panel setup

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_{t,post} & \mathbf{Y}_{c,post} \\ \mathbf{Y}_{t,pre} & \mathbf{Y}_{c,pre} \end{pmatrix} = \begin{pmatrix} \mathbf{Y}_{t,post}(1) & \mathbf{Y}_{c,post}(0) \\ \mathbf{Y}_{t,pre}(0) & \mathbf{Y}_{c,pre}(0) \end{pmatrix}$$

- Recall that our problem boils down to the estimate of an untreated "synthetic" unit
- Following Doudchenko and Imbens (2018), note estimators of the following form:

$$\hat{Y}_{t,post}(0) = \mu + \sum_{i \in c} \omega_i Y_{i,T}$$

  - A constant μ allows for very different averages (common in diff-in-diff)
  - Weights are allowed to vary across i – a simple average would be diff-in-diff

- We can now consider deviations from diff-in-diff

The synthetic control method (Abadie et al. (2010)

$$\hat{Y}_{t,post}(0) = \mu + \sum_{i \in c} \omega_i Y_{i,T}$$

- In ADH, they impose
  1. $\mu = 0$
  2. $\sum_i \omega_i = 1$
  3. $\omega_i \geq 0 \ \forall i$

- These three restrictions create a counterfactual California whose outcomes are within the support of the other states, and is a weighted sum of a subset of states

The synthetic control method (Abadie et al. (2010)

$$\hat{Y}_{t,post}(0) = \mu + \sum_{i \in c} \omega_i Y_{i,T}$$

- Formally, the ω_i need to be estimated, and are constructed by minimizing the distance between covariates in the pre-period:

$$||\boldsymbol{X}_{\text{treat}} - \boldsymbol{X}_{\text{control}} \boldsymbol{W}||$$

- The crucial piece tying this together: $\boldsymbol{X}$ can include both lagged outcomes, and covariates.

- Note we can now re-envision our panel data:
  - Observed outcomes: $\mathbf{Y}_{t,post}(1)$, $\mathbf{Y}_{c,post}(0)$
  - Observed covariates / predictors: $\mathbf{Y}_{t,pre}(0)$, $\mathbf{Y}_{c,pre}(0)$, $\mathbf{X}_t$, $\mathbf{X}_c$

- In many ways, this is just a matching problem using many characteristics!

The synthetic control method (Abadie et al. (2010)

$$\hat{Y}_{t,post}(0) = \mu + \sum_{i \in c} \omega_i Y_{i,T}$$

- Formally, the ω_i need to be estimated, and are constructed by minimizing the distance between covariates in the pre-period:

$$\{\hat{\omega}\}_i = \arg\min_{\boldsymbol{W}} ||\boldsymbol{X}_{\text{treat}} - \boldsymbol{X}_{\text{control}} \boldsymbol{W}||$$

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- In many ways, this is just a matching problem using many characteristics!

The synthetic control method (Abadie et al. (2010)

- This approach can be incredibly successful

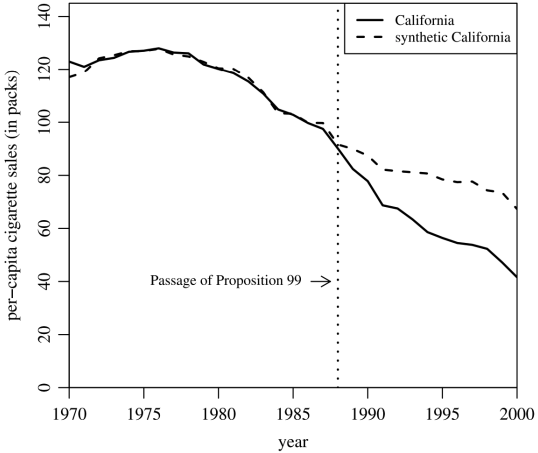

Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.

The synthetic control method (Abadie et al. (2010)

- This approach can be incredibly successful
- By careful construction of a synthetic control, can calculate counterfactual impacts due to policy
- Still subject to same caveats from DinD – not invariant to some transformations (e.g. log and linear)

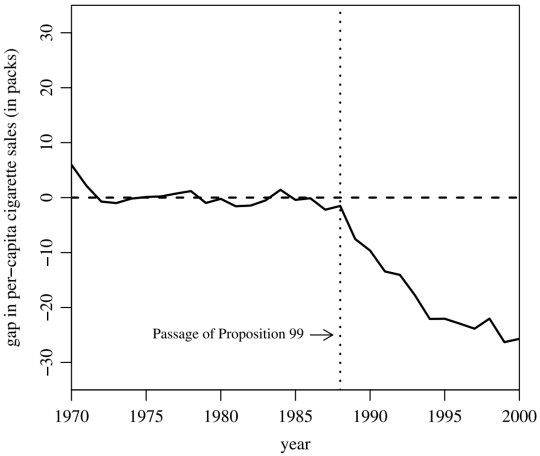

Figure 3. Per-capita cigarette sales gap between California and synthetic California.

Inference in the synthetic control method (Abadie et al. (2010)

- Inference for this method is slightly more complex, as there is only a single treated unit
  - Large sample asymptotics unlikely to work
- Placebo approach is standard: apply method to each potential control unit, and report effect in period
- Analogy here is to a randomization inference argument, comparing to a "null" effect

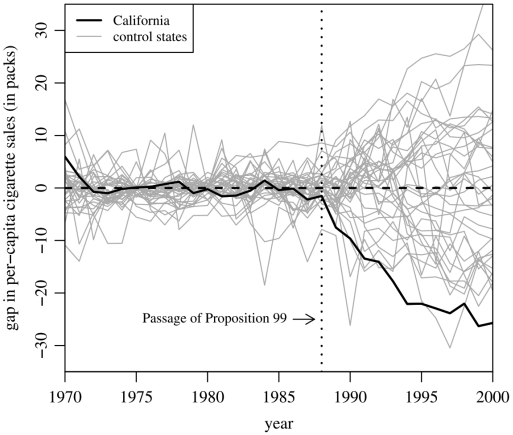

Figure 5. Per-capita cigarette sales gaps in California and placebo gaps in 34 control states (discards states with pre-Proposition 99 MSPE twenty times higher than California's).

Synthetic Diff-in-diff

- In Arkhangelsky et al. (2019), they show you can rewrite the synthetic control estimator as

$$(\hat{\mu}, \hat{\gamma}, \hat{\tau}) = \arg\min_{\mu,\gamma,\tau} \sum_i \sum_t (Y_{it} - \mu - \gamma_t - D_{it}\tau)^2 \hat{\omega}_i,$$

  subject to the $\hat{\omega}_i$ chosen via the SC approach

- Contrast that with DID:

$$(\hat{\mu}, \hat{\alpha}, \hat{\gamma}, \hat{\tau}) = \arg\min_{\mu,\gamma,\tau} \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \gamma_t - D_{it}\tau)^2$$

- They then propose a more robust approach, called Synthetic diff-in-diff, which estimates

$$(\hat{\mu}, \hat{\alpha}, \hat{\gamma}, \hat{\tau}) = \arg\min_{\mu,\gamma,\tau} \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \gamma_t - D_{it}\tau)^2 \hat{\omega}_i \hat{\lambda}_t$$

- This approach relaxes the parallel trends assumption by requiring parallel trends in an underlying approximate factor structure

Synthetic Diff-in-diff

- Key difference is twofold:
  1. Pre-trend means do not need to match "exactly"
  2. Weighting is not equivalent across all time periods

- Conceptually – different ways to generate the counterfactual given a model

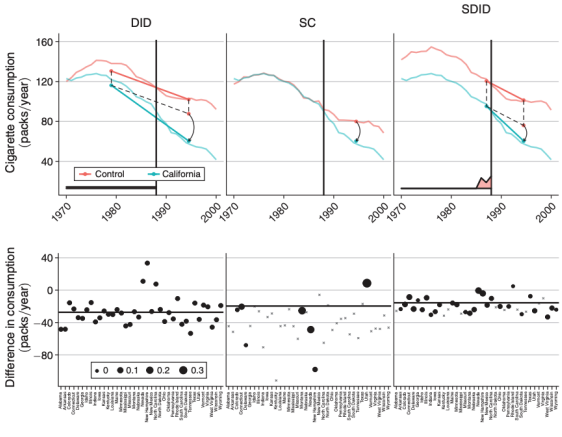

FIGURE 1. A COMPARISON BETWEEN DID, SC, AND SDID ESTIMATES FOR THE EFFECT OF CALIFORNIA PROPOSITION 99 ON PER-CAPITA ANNUAL CIGARETTE CONSUMPTION (IN PACKS/YEAR)

Synthetic Diff-in-diff

- Key difference is twofold:
  1. Pre-trend means do not need to match "exactly"
  2. Weighting is not equivalent across all time periods

- Conceptually – different ways to generate the counterfactual given a model

$$\hat{\tau} = \hat{\delta}_{tr} - \sum_{i=1}^{N_{co}} \hat{\omega}_i \hat{\delta}_i \qquad \text{where} \qquad \hat{\delta}_{tr} = \frac{1}{N_{tr}} \sum_{i=N_{co}+1}^{N} \hat{\delta}_i .$$

$$\hat{\delta}_i^{sc} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it},$$

$$\hat{\delta}_i^{did} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it} - \frac{1}{T_{pre}} \sum_{t=1}^{T_{pre}} Y_{it},$$

$$\hat{\delta}_i^{sdid} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it} - \sum_{t=1}^{T_{pre}} \hat{\lambda}_t^{sdid} Y_{it}.$$

Synthetic Diff-in-diff

- So far, synth dind method discussion focused on single adoption period.
  1. Staggered adoption in synthetic control isn't meaningful
  2. How can you adopt it?

- Conceptually – split up the adoption timings a la Calloway & Sant'anna and others

$$\mathbf{W} = \begin{pmatrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 4 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 5 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 6 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Synthetic Diff-in-diff

- So far, synth dind method discussion focused on single adoption period.
  1. Staggered adoption in synthetic control isn't meaningful
  2. How can you adopt it?
- Conceptually – split up the adoption timings a la Calloway & Sant'anna and others

$$\mathbf{W}^1 = \begin{pmatrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 6 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix},$$

$$\mathbf{W}^2 = \begin{pmatrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 4 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

So what about synthetic methods?

- Both an old field, and a new one – lots of new methodological papers coming out
  - It is a very cool method!
- So far, limited application by researchers. Why?
- My thoughts:
  - These are strong structural assumptions, and not clear we have good tests yet
  - Despite concerns re: pre-trends in dind, the assumptions felt testable
- Researcher degrees of freedom seem multifold. True in DinD too, but perhaps more transparent?
  - More worrisome: dind is equally problematic, but we aren't aware of it
- If researchers are more willing to understand that DinD is sensitive to functional form, ML methods that construct counterfactual outcomes are a natural direction

My recommendation / takeaway

- Synthetic control is the ideal approach when faced with a single treatment
  - By far the most natural approach in this setting, and is a practical approach
  - Typcial approach – get a good synthetic control for a given treatment. If none exists, stop. Ben-Michael, Feller and Rothstein (2021) provide a better approach, which adjusts for imperfect pre-match.

- Synthetic DiD seems very promising as a generalization
  - Key question is convincing readers why this shoudl work better than traditional method
  - My view: empirical papers will first need to show how / why their method works with both diff-in-diff and synth diff-in-diff

- Key point: *all of this relies on a model of the control outcome*

- Three packages to explore: augsynth/tidysynth andsynthdid packages (original synth package is tough to use)

Taking a step back on design

- The previous discussion all sits on the "model-based" approach for difference-in-differences
  - E.g. Correctly modeling $E(Y(0))$ (the counterfactual)
  - Challenging when thinking about logs vs. levels, for example

- Alternative approach: random timing?

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Design-based analysis in Difference-In-Differences settings with staggered adoption ☆

Susan Athey [a, b], Guido W. Imbens [a, b, c, d]

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Abstract

In this paper we study estimation of and inference for average treatment effects in a setting with panel data. We focus on the staggered adoption setting where units, *e.g*, individuals, firms, or states, adopt the policy or treatment of interest at a particular point in time, and then remain exposed to this treatment at all times afterwards. We take a design perspective where we investigate the properties of estimators and procedures given assumptions on the assignment process. We show that under random assignment of the adoption date the standard Difference-In-Differences (DID) estimator is an unbiased estimator of a particular weighted average causal effect. We characterize the exact finite sample properties of this estimand, and show that the standard variance estimator is conservative.

Taking a step back on design

- The previous discussion all sits on the "model-based" approach for difference-in-differences
  - E.g. Correctly modeling $E(Y(0))$ (the counterfactual)
  - Challenging when thinking about logs vs. levels, for example
- Alternative approach: random timing?
- Key point: we can define a propensity score where the treatment is considered "random"

Define $W : \mathbb{A} \times \mathbb{T} \mapsto \{0, 1\}$, with $W(a, t) = \mathbf{1}_{a \leq t}$ to be the binary indicator for the adoption date a preceding t, and define W_{it} to be the indicator for the policy having been adopted by unit i prior to, or at, time t, so that:

$$W_{it} \equiv W(A_i, t) = \mathbf{1}_{A_i \leq t}.$$

The $N \times T$ matrix $\boldsymbol{W}$ with typical element W_{it} has the form:

$$\boldsymbol{W}_{N \times T} = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & T & \text{(time period)} \\ & & & & & & \\ 0 & 0 & 0 & 0 & \dots & 0 & \text{(never adopter)} \\ 0 & 0 & 0 & 0 & \dots & 1 & \text{(late adopter)} \\ 0 & 0 & 0 & 0 & \dots & 1 & \\ 0 & 0 & 1 & 1 & \dots & 1 & \\ 0 & 0 & 1 & 1 & \dots & 1 & \text{(medium adopter)} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 1 & 1 & 1 & \dots & 1 & \text{(early adopter)} \end{pmatrix}.$$

Taking a step back on design

- In the panel setting, it's worth considering whether you can just consider the policy "as-if" random, conditional on some controls

- Can you make units sufficiently comparable?
  - E.g. if all firms greater than 100 employees are treated, could you compare firms just below 100 and just above 100 (to come in RD!)
  - This would *not* require a functional form assumption

- Are there controls such that the groups appear "observationally identical" in the pre-period?
  - Synthetic control + propensity score methods (to come)

Random Timing in Diff-in-Diff

- Athey and Imbens (2022) take design-based approach to diff-in-diff
  - What does this mean? Can consider a well-defined propensity scores for the treatment
  - Focus on staggered adoption with binary treatment
  - Since treamtent is absorbing, this simplifies problem into "when did I get the treatment (if ever)?"
- E.g. define a propensity score: $Pr(A_i = a)$ over when the assignment occurs.
- Key question: who is the relevant counterfactual group?

Three key assumptions (not testable!)

1. Random assignment (can make this happen by design)
2. No anticipation (we assume this already)
3. Invariance to history ($1_{a \leq t}(Y_{it}(a) - Y_{it}(1)) = 0$) (no causal effect of an early adotpion vs. a later adoption on outcome as long as adoption occured before or on period t)

Efficiency in estimating staggered random roll-out

- Roth and Sant'anna (2023) show that it is much more efficient to condition on lagged outcomes over using standard diff-in-diff in staggered roll-outs

- E.g. Consider the class of estimators:

$$\hat{\theta}_\beta = (\overline{Y}_{22} - \overline{Y}_{2\infty}) - \beta(\overline{Y}_{12} - \overline{Y}_{1\infty}) \tag{1}$$

  Did is the special case of $\beta = 1$.

- We want to put more weight on this setting when the lagged outcomes are more predictive, and less when it is not!
  - Intuition is similar to synthetic controls

Covariates

- With time-invariant covariates that treatment does not affect, controlling for covariates is conditional parallel trends assumption

$$E(Y_{i,2}(0) - Y_{i,1}(0)|D_i = 1, X_i) = E(Y_{i,2}(0) - Y_{i,1}(0)|D_i = 0, X_i) \tag{2}$$

- However, this isn't enough to satisfy this TWFE regression approach in two periods:

$$Y_{it} = \alpha_i + \phi_t + 1(t = 2)D_i\beta + X_i 1(t = 2)\gamma + \epsilon_{it} \tag{3}$$

Why? (consider age)

- Even worse, if covariates affected by treatment, this is a form of collider bias!
  - Should be very careful about time-varying covariates

Multiple Treatments in Diff-in-diff

- Hull (2018) and deChaisemartin and D'hautefeuille (2022) are relevant cites

- One "easy" context that this shows up (Hull 2018) is "mover" designs
  - I move from city i to city j – if my move was random, can we use it to understand the effect of city j?
  - Without strong additional assumptions, challenging to interpret coefficients from standard TWFE

$$Y_{it} = \alpha_i + \tau_t + \sum_{j \neq 0} \beta_j D_{ijt} + \epsilon_{it} \tag{4}$$

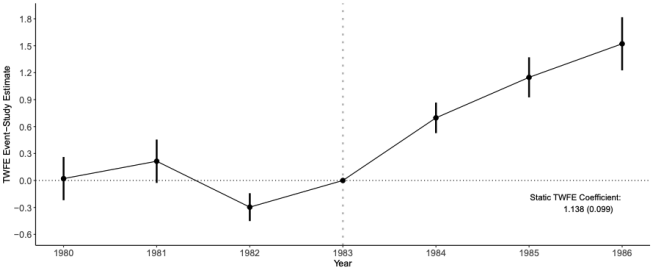
- Consider 2 period case, and use first differences:

$$\Delta Y_i = \tau + \sum_{j \neq 0} \beta_j \Delta D_{ij} + \Delta \epsilon_i \tag{5}$$

- Recall Goldsmith-Pinkham, Hull and Kolesar (2022) – multiple treatments (ΔD_{ij}) is potentially contaminated and negative weighted.

Fix ideas with example (Callaway, Goodman-Bacon, Sant'anna (2024))

Figure 1: Two-Way Fixed Effects Event-Study Estimates of the Effect of Medicare's Reimbursement Reform on Hospital Input Mix



Notes: The figure plots TWFE event-study coefficients and their 95% confidence intervals from regressions with hospital fixed effects, year fixed effects, and the 1983 Medicare inpatient share (m_i) interacted with either a dummy for years after 1983 or the year dummies. The outcome variable is the depreciation share of total operating expenses, a measure of hospitals' capital/labor ratio. The data cover the years 1980-1986 and come from the American Hospital Association's annual survey (American Hospital Association, 1986). We dropped 860 hospitals (out of 6741) that have missing data for the outcome. We also report the static TWFE coefficient and standard errors associated with (1.1). All standard errors are clustered at the hospital level.

$$y_{it} = \theta_t + \eta_i + \beta D_i \cdot Post_t + \varepsilon_{it}$$

In 1983, Prospective Payment System changed payment for Medicare patients to hospitals

Continuous treatment

- Recall that estimating treatment effects with random assignment and continuous treatments was data hungry, but doable

- Without random assignment, and unconstrained heterogeneity, the challenge is more complicated

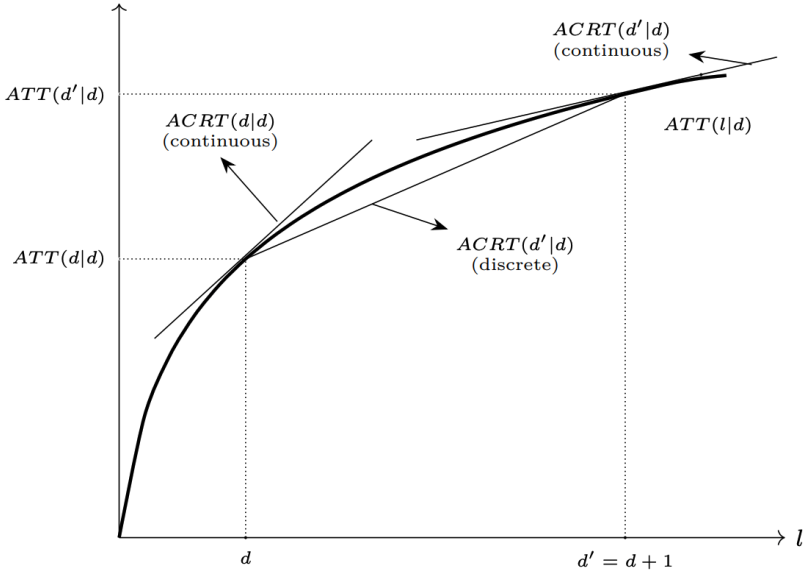
- First, consider what the estimand of interest is. Note that with potential outcomes, we now have $Y_i(D_i)$ with D_i multivalued.

- So what is the contrast of interest?

$$ATT(d|d) = E(Y_i(d) - Y_i(0)|D_i = d) \tag{6}$$

$$ACRT(d|d) = \left.\frac{\partial E(Y_i(l)|D_i = d)}{\partial l}\right|_{l=d} \tag{7}$$

ACRT vs. ATT


$ACRT(d'|d)$
(continuous)
$ATT(d'|d)$
$ACRT(d|d)$
(continuous)
$ATT(l|d)$
$ACRT(d'|d)$
(discrete)
$ATT(d|d)$
l
d
$d' = d + 1$

Continuous treatment – Estimand

- How do these differ? Consider their interpretations:
  - ATT: effect relative to baseline of zero effect
  - ACRT: effect of marginally increasing treatment, evaluted at a given point (can integrate too!)

- Under parallel trends, what can we identify?
  - First, parallel in what counterfactual? Under *no* treatment

$$E(Y_2(0) - Y_1(0)|D = d) = E(Y_2(0) - Y_1(0)|D = 0). \tag{8}$$

  implies

$$ATT(d|d) = E(\Delta Y|D = d) - E(\Delta Y|D = 0). \tag{9}$$

- However, this does not identify anything about the ACRT

Continuous treatment – Estimand

- Can we move between ATT and ACRT?

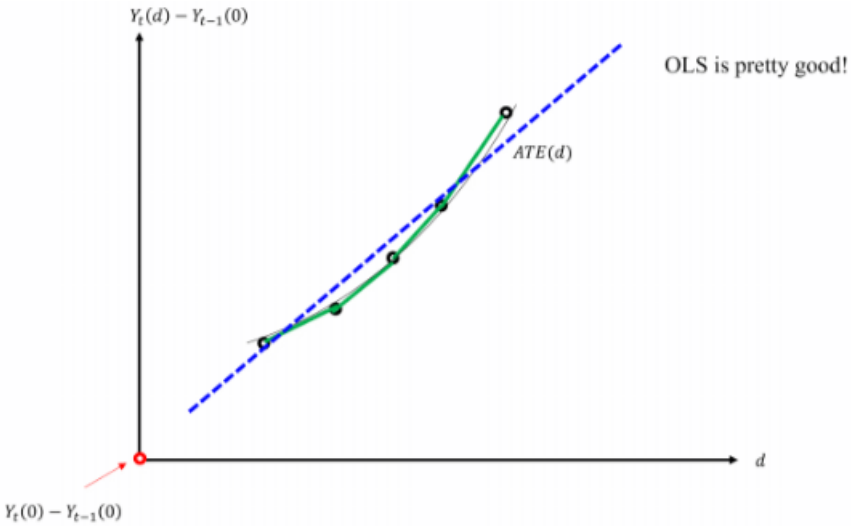
- Under standard parallel trends,

$$\frac{\partial E(\Delta Y|D=d)}{\partial d} = \frac{\partial ATT(d|d)}{\partial d} = ACRT(d|d) + \frac{\partial ATT(d|l)}{\partial l}|_{l=d} \tag{10}$$

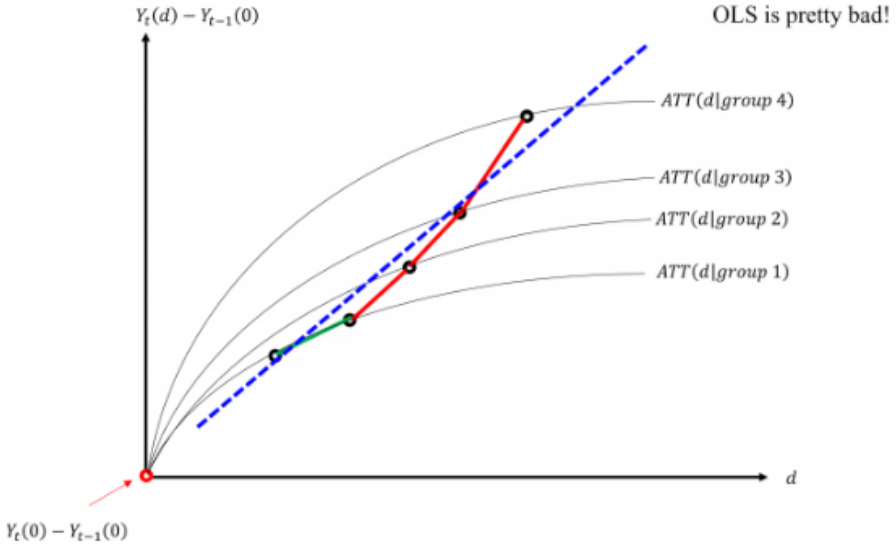
- Consider comparing two ATTs:

$$\begin{aligned} ATT(d|d) - ATT(d'|d') &= (E[\Delta Y|D=d] - E[\Delta Y|D=0]) \\ &- (E[\Delta Y|D=d'] - E[\Delta Y|D=0]) \\ &= E[\Delta Y|D=d] - E[\Delta Y|D=d'] \\ &= E[\Delta Y(d) - \Delta Y(d')|D=d] + ATT(d'|d) - ATT(d'|d') \end{aligned}$$

Continuous treatment – challenge



Continuous treatment – challenge



Continuous treatment – stronger assumption about heterogeneity

- "Strong" parallel trends assumption: $E(\Delta Y_i(d)) = E(\Delta Y_i(d)|D_i = d)$
- No selection bias "on average"
- Effectively assuming some kind of homogeneity in the treatment across groups
- Under this set of assumptions, you can identify the ATE as well as the ATT (and the ACR along with the ACRT)

Continuous treatment – what if there is no untreated group?

- Can consider parallel trends relative to some d_l "baseline" group
- However, this mixes the selection effect based on treatment:

$$E(\Delta Y|D_i = d) - E(\Delta Y|D_i = d_l) = ATT(d|d) - ATT(d_l|d_l) \tag{11}$$

and would need that $ATT(d_l|d) = ATT(d_l|d_l)$ to identify the ATT

So what does TWFE do?

Table 1: TWFE Decomposition Weights

Decomposition	$D > 0$ Weights	$D = 0$ Weights
Causal response	$w_1^{\text{acr}}(l) = \frac{(\mathbb{E}[D\|D \geq l] - \mathbb{E}[D])\mathbb{P}(D \geq l)}{\text{Var}(D)}$	$w_0^{\text{acr}} = \frac{(\mathbb{E}[D\|D > 0] - \mathbb{E}[D])\mathbb{P}(D > 0)d_L}{\text{Var}(D)}$
Levels	$w_1^{\text{lev}}(l) = \frac{(l - \mathbb{E}[D])}{\text{Var}(D)} f_D(l)$	$w_0^{lev} = -\frac{\mathbb{E}[D]\mathbb{P}(D = 0)}{\text{Var}(D)}$
Scaled levels	$w^{\text{s}}(l) = l\frac{(l - \mathbb{E}[D])}{\text{Var}(D)} f_D(l)$	
Scaled 2×2	$w_1^{2\times 2}(l, h) = \frac{(h - l)^2 f_D(h) f_D(l)}{\text{Var}(D)}$	$w_0^{2\times 2}(h) = \frac{h^2 f_D(h)\mathbb{P}(D = 0)}{\text{Var}(D)}$

Notes: The table provides the formulas for the weights used in the decompositions of β^{twfe} provided in this section.

Theorem 3.4. *Under Assumptions 1, 2(a), 3, and 4, β^{twfe} can be decomposed in the following ways:*

(a) Causal Response Decomposition:

$$\beta^{twfe} = \int_{d_L}^{d_U} w_1^{acr}(l) \left(ACRT(l|l) + \underbrace{\frac{\partial ATT(l|h)}{\partial h}\Big|_{h=l}}_{\text{selection bias}} \right) dl + w_0^{acr} \frac{ATT(d_L|d_L)}{d_L}$$

where the weights are always positive and integrate to 1.

So what does TWFE do?

Table 1: TWFE Decomposition Weights

Decomposition	$D > 0$ Weights	$D = 0$ Weights
Causal response	$w_1^{\text{acr}}(l) = \frac{(\mathbb{E}[D\|D \geq l] - \mathbb{E}[D])\mathbb{P}(D \geq l)}{\text{Var}(D)}$	$w_0^{\text{acr}} = \frac{(\mathbb{E}[D\|D > 0] - \mathbb{E}[D])\mathbb{P}(D > 0)d_L}{\text{Var}(D)}$
Levels	$w_1^{\text{lev}}(l) = \frac{(l - \mathbb{E}[D])}{\text{Var}(D)} f_D(l)$	$w_0^{lev} = -\frac{\mathbb{E}[D]\mathbb{P}(D = 0)}{\text{Var}(D)}$
Scaled levels	$w^{\text{s}}(l) = l\frac{(l - \mathbb{E}[D])}{\text{Var}(D)} f_D(l)$	
Scaled 2×2	$w_1^{2\times 2}(l, h) = \frac{(h - l)^2 f_D(h) f_D(l)}{\text{Var}(D)}$	$w_0^{2\times 2}(h) = \frac{h^2 f_D(h)\mathbb{P}(D = 0)}{\text{Var}(D)}$

Notes: The table provides the formulas for the weights used in the decompositions of β^{twfe} provided in this section.

(b) Levels Decomposition:

$$\beta^{twfe} = \int_{d_L}^{d_U} w_1^{lev}(l) ATT(l|l)\, dl,$$

where $w_1^{lev}(l) \lesseqgtr 0$ *for* $l \lesseqgtr \mathbb{E}[D]$, *and* $\int_{d_L}^{d_U} w_1^{lev}(l)\, dl + w_0^{lev} = 0$.

(c) Scaled Levels Decomposition:

$$\beta^{twfe} = \int_{d_L}^{d_U} w^s(l) \frac{ATT(l|l)}{l}\, dl,$$

where $w^s(l) \lesseqgtr 0$ *for* $l \lesseqgtr \mathbb{E}[D]$, *and* $\int_{d_L}^{d_U} w^s(l)\, dl = 1$.

Under strong parallel trends, TWFE gets convex ACR

Theorem 3.4. *Under Assumptions 1, 2(a), 3, and 4, β^{twfe} can be decomposed in the following ways:*

(a) Causal Response Decomposition:

$$\beta^{twfe} = \int_{d_L}^{d_U} w_1^{acr}(l) \left(ACRT(l|l) + \underbrace{\frac{\partial ATT(l|h)}{\partial h}\Big|_{h=l}}_{\text{selection bias}} \right) dl + w_0^{acr} \frac{ATT(d_L|d_L)}{d_L}$$

where the weights are always positive and integrate to 1.

- Under strong parallel trends, the selection bias terms disappear (and with $d_l = 0$, last term disappears in (a))
- Weights are also convex and weakly causal! However, don't necessary map to an estimand of interest. Related to OLS weights from previous discussions
- ATT composition does not have convexity property

So what can you do?

- Easiest thing is to use untreated units as baseline category

- When treatments are discrete:

$$\Delta Y_i = \beta_0 + \sum_j 1(D_i = d_j)\beta_j + \epsilon_i \tag{12}$$

  Under parallel trends, each β is $ATT(d|d)$, and with strong parallel trends, $\beta_j - \beta_{j-1}$ is $ACR(d_j)$.

- Life is more complicated when d is continuous of the groups are quite small at a given treatment:

$$\Delta Y_i = \sum_k \Psi_{Kk}(D)\beta_{Kk} + \epsilon_i, \tag{13}$$

  where $\Psi_{Kk}(D)$ is a basis function for the treatment (implementation can be found in paper)

Checklist

- First, consider the treatment features of your setup
  - What is your treatment? Binary or continuous?
  - Single or multiple?
  - Absorbing or not?
  - How many treated units? Single or many?
  - How many interventions? Single or staggered timing?
- Second, consider the relevant identifying assumptions for your condition
- Third, what tests are feasible in your setting?
- Fourth, what estimates do you want?
- Fifth, what relaxations of assumptions can you make?

Treatment features

- If treatment is binary or only a few values, easy to consider the simple case
  - Particularly helps if it's absorbing!
- If continuous and/or not absorbing, need to worry about many more homogeneity assumptions
- First order of business – *define your estimand*.
  - What treatment do you care about?
- Can you simplify the problem in some way if it's more complex?

Assumptions

- Write out carefully in math and then words what assumptions you need
  - Parallel trends (with or without covariates)?
  - No anticipation?
  - Others?

- If you know your estimand, much easier to know what you need

- Do you need to construct your estimator using lagged outcomes, e.g. synthetic control?

Testable assumptions

- What is testable? Pre-trends can be useful, but underpowered
- If using random timing, not much is testable
  - But could look for balance across timing groups in exogeneous features

What estimates do you want?

- Do you need a long run effect? Short-run?
  - Long-run effect relies *heavily* on parametric extrapolation. Is it plausible?
- Are you studying ATTs? CATTs? ACRTs?

Relaxation of assumptions?

- What group comparaisons are you making for parallel trends?
- Could you assume random timing instead of parallel trends?
- Could you condition on lagged outcomes?

Conclusion

- Difference in difference is hugely powerful in applied settings
- Does not require random assignment, but rather implementation of policies that differentially impacts different groups and is not confounded by other shocks at the same time.
- Can be a great application of big data, with convincing graphs that highlight your application
- Also allows for partial tests of identifying assumptions
- Worth carefully thinking about what your identifying assumptions are in each setting, and transparently highlighting them.
- Important to note that this always identifies a *relative* effect, and to aggregate, you will typically need a model and additional strong assumptions (see Auclert, Dobbie and Goldsmith-Pinkham (2019) for an example in a macro setting).

My takeaways from new literature

- Beware weak tests of pre-trends. Consider using R&R's partial identification tests to assess robustness of results.
- Do not worry about the new literature on staggered timings if you only have one rollout period!
- Think carefully about your estimand if you're using a staggered timing DinD – what's your counterfactual in each case?
  - Software exists for many of these papers. This is doable!
- When plotting confidence intervals in event studies, you should plot uniform confidence intervals.